Activity		Estimated duration (weeks)		
i	j	Optimistic	Most likely	Pessimistic
1	2	1	1	7
1	3	1	4	7
1	4	2	2	8
2	5	1	1	1
3	5	2	5	14
4	6	2	5	8
5	6	3	6	15

(i) Draw the project network

(ii) Find the expected duration and variance of each activity. What is the expected project length?

(iii) Calculate the variance and standard deviation of project length. What is the probability that the project will be completed.

 Roll No. .....

521352(14)

M.C.A (Third Semester) Examination, April-May 2017

(New Scheme)

(Computer Application Branch)

COMPUTER ORIENTED OPTIMIZATION

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 40

Note : *Part (a) of each question is compuslory and carries 4 marks attempt any* ***two*** *parts from (b), (c) and (d) of each question which carrying 8 marks each.*

Unit-I

1. (a) Establish the difference between :

(i) Feasible solution

(ii) Basic feasible solution.

(b) Use the graphical method to solve the following LPP :

Minimize $Z = -2x_1 + 2x_2$

Subject to the constraints :

$$-x_1 + 3x_2 \le 10, \; x_1 + x_2 \le 6$$

$$x_1 - x_2 \le 2 \text{ and } x_1 \ge 0, \; x_2 \ge 0.$$

(c) Use Simplex method to :

Minimize $Z = x_2 - 3x_3 + 2x_3$

Subject to the constraints :

$$3x_2 - x_3 + 2x_5 \le 7$$

$$-2x_2 + 4x_3 \le 12$$

$$-4x_2 + 3x_3 + 8x_5 \le 10; \; x_2 \ge 0, \; x_3 \ge 0$$

and $x_5 \ge 0$.

(d) Use Big M method to :

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

Subject to the constraints :

$$x_1 + 2x_2 + 3x_3 = 15$$

$$2x_1 + x_2 + 5x_3 = 20$$

$$x_1 + 2x_2 + x_3 + x_4 = 10$$

$$x_1, x_2, x_3, x_4 \ge 0.$$

Unit-II

2. (a) What is an assignment problem? Explain.

(b) Use Vogel's approximation method to obtain an initial basic feasible solution of the transportation problem :

	D	E	F	G	Available
A	11	13	17	14	250
B	16	18	14	10	300
C	21	24	13	10	400
Demand	200	225	275	250	

(c) A company wishes to assign 3 jobs to 3 machines in such a way that each job is assigned to some machine and no machine works on more than one job. The cost of assigning job i to machine j is given by the

matrix below :

$$\text{Costmatrix} : \begin{bmatrix} 8 & 7 & 6 \\ 5 & 7 & 8 \\ 6 & 8 & 7 \end{bmatrix}$$

Draw the associated network. Formulate the network LPP and find the minimum cost of making the assignment.

(d) A machine operator processes five types of items on his machine each week and must choose a sequence for them. The setup cost per change depends on the item presently on the machine and the set-up to be made according to the following table :

From item	To item				
	A	B	C	D	E
A	∞	4	7	3	4
B	4	∞	6	3	4
C	7	6	∞	7	5
D	3	3	7	∞	7
E	4	4	5	7	∞

If he processes each type of item once and only once each week, how should be sequence the items on his machine in order to minimize the total setup cost?

Unit-III

3. (a) Describe the basic characteristics of an inventory system.

(b) A dealer supplies you the following information with regard to a product dealt in by him :

Annual demand : 10,000 units; ordering cost : Rs.10 per order ; Price : Rs. 20 per units Inventory carrying cost : 20% of the value of inventory per year.

The dealer is considering the possibility of allowing some back-order to occur. He has estimated that the annual cost of backordering will be 25% of the value of inventory :

(i) What should be the optimum number of units of the product he should buy in one lot?

(ii) What quantity of the product should be allowed to be back-order. 1 if any?

(iii) What would be the maximum quantity of inventory at any time of the year?

(iv) Would you recommend to allow back ordering? If so what would be the annual cost saving by adopting the policy of back ordering.

(c) A small shop produces three machine parts I, II and III in lots. The shop has limited storage space sufficient only for 500 units of all type of items. The relevant data for the three items is given below :

Item	I	II	III
Demand rate (unit / month)	600	1200	1500
Cost per unit (Rs.)	5	10	15
Setup cost per lot (Rs)	100	50	200

The inventory carrying charges for the shop are 20% of the average valuation per month for each item. If stock outs are not allowed, determine the optimum lot size for each item.

(d) The probability distribution of monthly sales of certain item is as follows :

Monthly sales	0	1	2	3	4	5	6
Probability	0·02	0·05	0·3	0·27	0·2	0·1	0·06

The cost of carrying inventory is Rs. 10 per unit per month. The current policy is to maintain a stock of four items at the beginning of each month. Assuming that the cost of shortage is proportional to both time and quantity short, obtain the inputed cost of a shortage of one item for one unit of time.

Unit-IV

4. (a) Give an example of a sequencing model.

(b) In a factory, there are six jobs to perform each of which should go through two machines A and B, in the order A, B. The processing timing for the jobs are given here. You are required to determine the sequence for performing the jobs that would minimize the total elapsed time T. What is the value of T?

Job	J_1	J_2	J_3	J_4	J_5	J_6
Machine A :	1	3	8	5	6	3
Machine B :	5	6	3	2	2	10

(c) Determine the optimal sequence of jobs that minimize the total elapsed time based on the following information processing time on machine is given in hours and passing is not allowed :

Job	A	B	C	D	E	F	G
M_1	3	8	7	4	9	8	7
M_2	4	3	2	5	1	4	3
M_3	6	7	5	11	5	6	12

(d) In a railway marshalling yard, goods trains arrive at a rate of 30 trams per day. Assuming that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average 36 minutes. Calculate the following :

(i) the mean queue size, and

(ii) the probability that the queue size exceeds 10.

If the input of trams increased to an average 33 per day. What will be the change is (i) and (ii) ?

Unit-V

5. (a) State the objectives, advantages and disadvantages of network analysis.

(b) Construct the network diagram comprising activities $B, C, \ldots\ldots, Q$ and N such that the following constraints are satisfied :

$B < E, F;\ C < G, L;\ E, G < H;\ L, H < I;\ L < M;$

$H < N;\ H < J;\ I\ ;\ J < P;\ P < Q$

The notation X < Y means that the activity X must be finished before Y can begin.

(c) A small project consists of seven activities for which the relevant data are givin below :

Activity	A	B	C	D	E	F	G
Preceding activity	–	–	–	A,B	A,B	C,D,E	C,D,E
Activity duration (days)	4	7	6	5	7	6	5

(i) Draw the network and find the project completion time.

(ii) Calculate total float for each of the activities and highlight the critical path.

(iii) Draw the time scaled diagram.

(d) A small project is composed of seven activities whose time estimates are listed in the table as follows :